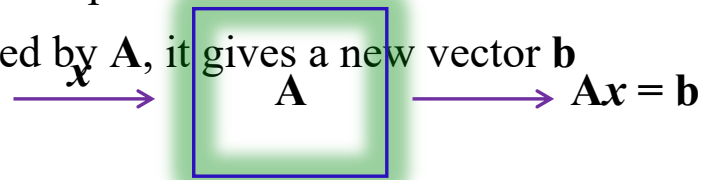


Eigen Values and Eigen Vectors

Eigenvalues and Eigenvectors

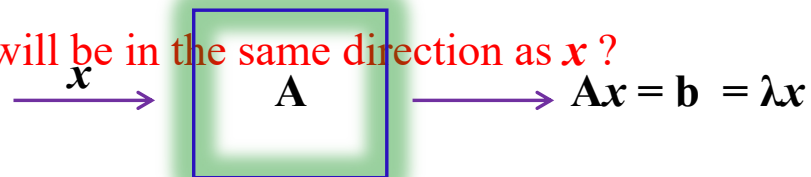
- Defined only for square matrices.
- A system of equations can be expressed as $\mathbf{Ax} = \mathbf{b}$

When a vector $\mathbf{x}$ is multiplied by $\mathbf{A}$, it gives a new vector $\mathbf{b}$



- This new vector $\mathbf{b}$, could be in any direction and it depends upon the vector $\mathbf{x}$

- Which $\mathbf{x}$ will give a $\mathbf{b}$ that will be in the same direction as $\mathbf{x}$?

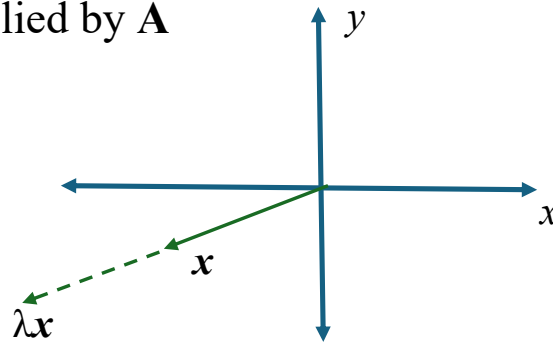


Eigenvalues and Eigenvectors

- Certain exceptional vectors $\mathbf{x}$ are in the same direction as $\mathbf{Ax}$ and those are called as the 'eigenvectors'

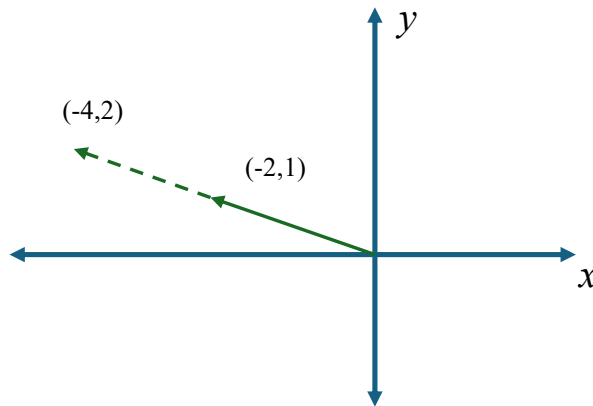
i.e. it may be written as $\mathbf{Ax} = \lambda\mathbf{x}$. The number λ is an eigenvalue of $\mathbf{A}$

- The eigenvalue λ tells whether the special vector $\mathbf{x}$ is stretched or shrunk or reversed or left unchanged when it is multiplied by $\mathbf{A}$



Eigenvalues and Eigenvectors

- $\mathbf{Ax} = \lambda\mathbf{x}$ where $\mathbf{A} = \begin{bmatrix} -2 & 1 \\ 1 & 4 \end{bmatrix}$ and $\mathbf{x}$ is a multiple of $\begin{bmatrix} -2 \\ 1 \end{bmatrix}$
- Take $\mathbf{x} = \begin{bmatrix} -4 \\ 2 \end{bmatrix}$; when $\mathbf{A}$ is multiplied with $\mathbf{x}$, we get $\mathbf{Ax} =$

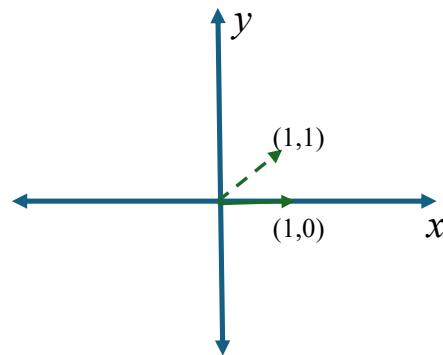


- The direction of the vector $\mathbf{x}$ hasn't changed even on multiplication with the matrix $\mathbf{A}$. Such a vector $\mathbf{x}$ is called as the eigenvector of the matrix $\mathbf{A}$

Eigenvalues and Eigenvectors

- Consider some vector $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$. Let us see what happens when this vector is multiplied by $\mathbf{A}$

$$\mathbf{A} = \begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}, \mathbf{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \mathbf{Ax} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$



- Clearly, the vectors $\mathbf{x}$ and $\mathbf{Ax}$ are not in the same direction. Hence, it is not an eigenvector

Eigenvalues and Eigenvectors

- The eigenvalue λ could be zero. Then $\mathbf{A}\mathbf{x} = 0\mathbf{x}$ means that this eigenvector $\mathbf{x}$ is in the null space

- If $\mathbf{A}$ is the identity matrix,

$$\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \lambda \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \Rightarrow \lambda = 1$$

all vectors are eigenvectors of $\mathbf{I}$ and all eigenvalues are $\lambda = 1$

Eigenvalues and Eigenvectors

$$\begin{bmatrix} 1 & -2 \\ 1 & 4 \end{bmatrix}$$

Example-1:

- Consider a matrix $A =$

- To find eigenvalue: $\det(A - \lambda I) = 0$

$$\det(A - \lambda I) = 0$$

$$(1 - \lambda)(4 - \lambda) + 2 = 0$$

On solving, we get,

$$\lambda = 3, 2$$

Eigenvalues and Eigenvectors

- Eigenvalues: $\lambda = 3, 2$

- When $\lambda = 3$,

$$\begin{bmatrix} -2 & -2 \\ 1 & 1 \end{bmatrix} \xrightarrow{\begin{array}{c} \text{R}_1 \leftrightarrow \text{R}_2 \\ \text{R}_2 = \text{R}_2 + 2\text{R}_1 \end{array}} \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix}$$

$$A - 3I =$$

- So, we have the equation –

$$x_1 + x_2 = 0 \Rightarrow x_1 = -x_2 \quad \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

- Eigenvector x is any multiple of

Eigenvalues and Eigenvectors

- when $\lambda=2$,

$$A-2I = \begin{bmatrix} -1 & -2 \\ 1 & 2 \end{bmatrix} \xrightarrow{\begin{array}{c} \text{R}_1 \leftrightarrow \text{R}_2 \\ \text{R}_2 = \text{R}_2 + \text{R}_1 \end{array}} \begin{bmatrix} 1 & 2 \\ 0 & 0 \end{bmatrix}$$

So, we have the equation –

$$x_1 + 2x_2 = 0 \Rightarrow x_1 = -2x_2 \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

- Eigenvector x is any multiple of
- So the eigenvalues are 3, 2 and corresponding eigenvectors are

$$a \begin{bmatrix} -1 \\ 1 \end{bmatrix} \text{ and } b \begin{bmatrix} -2 \\ 1 \end{bmatrix}, a, b \neq 0$$

Eigenvalues and Eigenvectors

Example-2:

- To find the eigenvalues and eigenvectors of the matrix

$$A = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix}$$

- We first find the eigenvalue λ which satisfies the characteristic equation of the matrix A,

$$\det (A - \lambda I) = 0$$

$$A - \lambda I = \begin{bmatrix} 1 & -3 & 3 \\ 3 & -5 & 3 \\ 6 & -6 & 4 \end{bmatrix} - \begin{bmatrix} \lambda & 0 & 0 \\ 0 & \lambda & 0 \\ 0 & 0 & \lambda \end{bmatrix} = \begin{bmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{bmatrix}$$

Eigenvalues and Eigenvectors

- Let us calculate $\det (A - \lambda I)$ now $\begin{vmatrix} -5-\lambda & 3 \\ -6 & 4-\lambda \end{vmatrix} \quad \begin{vmatrix} 3 & 3 \\ 6 & 4-\lambda \end{vmatrix} \quad \begin{vmatrix} 3 & -5-\lambda \\ 6 & -6 \end{vmatrix}$

$$\det (A - \lambda I) = (1 - \lambda) \quad - (-3) \quad + 3$$

$$\det (A - \lambda I) = [(1 - \lambda) (-2 + \lambda + \lambda^2)] + [3(-6 - 3\lambda)] + [3(12 + 6\lambda)]$$

$$= 16 + 12\lambda - \lambda^3$$

- Therefore, $\det (A - \lambda I) = -\lambda^3 + 12\lambda + 16 = 0$

Eigenvalues and Eigenvectors

- To solve: $-\lambda^3 + 12\lambda + 16 = 0$

$$-\lambda^3 + 12\lambda + 16 = (\lambda - 4)(\lambda^2 + 4\lambda + 4)$$

- On solving $\lambda^2 + 4\lambda + 4$, we get $\lambda = -2$ (repeated root)
- Hence, the eigenvalues of A are $\lambda = 4, -2, -2$
- Once the eigenvalues of a matrix have been found, we can find the eigenvectors by Gaussian Elimination

Eigenvalues and Eigenvectors

- For each eigenvalue λ , we have

$$(A - \lambda I) x = 0$$

where x is the eigenvector associated with eigenvalue λ

- Case 1: when $\lambda=4$,

$$A - \lambda I =$$

$$A - 4I =$$

$$\begin{bmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{bmatrix}$$

$$\begin{bmatrix} -3 & -3 & 3 \\ 3 & -9 & 3 \\ 6 & -6 & 0 \end{bmatrix}$$

$$\left[\begin{array}{ccc|c} -3 & -3 & 3 & 0 \\ 3 & -9 & 3 & 0 \\ 6 & -6 & 0 & 0 \end{array} \right]$$

- Augmented matrix:

Eigenvalues and Eigenvectors

$$\begin{bmatrix} -3 & -3 & 3 & 0 \\ 3 & -9 & 3 & 0 \\ 6 & -6 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 = R_1 \times (-1/3)} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 3 & -9 & 3 & 0 \\ 6 & -6 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{\begin{matrix} R_2 = R_2 - (3R_1) \\ R_3 = R_3 - (6R_1) \end{matrix}} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & -12 & 6 & 0 \\ 0 & -12 & 6 & 0 \end{bmatrix}$$

$$\xrightarrow{R_2 = R_2 (-1/12)} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & -12 & 6 & 0 \end{bmatrix}$$

$$\xrightarrow{R_3 = R_3 + 12R_2} \begin{bmatrix} 1 & 1 & -1 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\xrightarrow{R_1 = R_1 - R_2} \begin{bmatrix} 1 & 0 & -1/2 & 0 \\ 0 & 1 & -1/2 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Eigenvalues and Eigenvectors

- So, we get the equations –

$$x_1 - \frac{1}{2} x_3 = 0$$

$$x_2 - \frac{1}{2} x_3 = 0$$

- So the eigenvector x is given by:

$$\begin{bmatrix} x_1 = x_3 / 2 \\ x_2 = x_3 / 2 \\ x_3 \end{bmatrix} \quad \begin{bmatrix} 1/2 \\ 1/2 \\ 1 \end{bmatrix}$$

$$x = \quad = x_3$$

Eigenvalues and Eigenvectors

- Case2: when $\lambda = -2$

$$A - \lambda I =$$

$$\begin{bmatrix} 1-\lambda & -3 & 3 \\ 3 & -5-\lambda & 3 \\ 6 & -6 & 4-\lambda \end{bmatrix}$$

$$A + 2I =$$

$$\begin{bmatrix} 3 & -3 & 3 \\ 3 & -3 & 3 \\ 6 & -6 & 6 \end{bmatrix}$$

- Augmented matrix:

$$\left[\begin{array}{ccc|c} 3 & -3 & 3 & 0 \\ 3 & -3 & 3 & 0 \\ 6 & -6 & 6 & 0 \end{array} \right]$$

Eigenvalues and Eigenvectors

$$\left[\begin{array}{ccc|c} 3 & -3 & 3 & 0 \\ 3 & -3 & 3 & 0 \\ 6 & -6 & 6 & 0 \end{array} \right] \xrightarrow{\boxed{R_1 = R_1 / 3}} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 3 & -3 & 3 & 0 \\ 6 & -6 & 6 & 0 \end{array} \right]$$

$$\xrightarrow{\boxed{\begin{array}{l} R_2 = R_2 - 3R_1 \\ R_3 = R_3 - 6R_1 \end{array}}} \left[\begin{array}{ccc|c} 1 & -1 & 1 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{array} \right]$$

So, we get the equations –

$$x_1 - x_2 + x_3 = 0$$

So the eigenvectors x associated with the eigenvalue $\lambda = -2$ are given by:

$$x = \begin{bmatrix} x_2 - x_3 \\ x_2 \\ x_3 \end{bmatrix} = x_3 \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} + x_2 \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Eigenvalues and Eigenvectors

- So the eigenvalues are $\lambda = 4, -2, -2$ and the eigenvectors are any multiple of

$$\begin{bmatrix} 1/2 \\ 1/2 \\ 1 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix} \text{ and } \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$$

Eigenvalue Decomposition

- Let $\mathbf{A}$ be an $n \times n$ matrix with n linearly independent eigenvectors then $\mathbf{A}$ can be factored into

$$\mathbf{A} = \mathbf{S} \mathbf{\Lambda} \mathbf{S}^{-1}$$

- The columns of $\mathbf{S}$ are eigenvectors of $\mathbf{A}$
- The diagonal matrix $\mathbf{\Lambda}$ has eigenvalues of $\mathbf{A}$

- With $\mathbf{A}$, $\mathbf{S}$ and $\mathbf{S}^{-1}$ available, we may find $\mathbf{\Lambda}$

$$\mathbf{\Lambda} = \mathbf{S}^{-1} \mathbf{A} \mathbf{S}$$

Eigenvalue Decomposition

Example:

$$\begin{bmatrix} 1 & -1 \\ -2 & 0 \end{bmatrix}$$

Consider a matrix $A =$

- To find eigenvalue: $\det(A - \lambda I) = 0$
 $\begin{vmatrix} 1-\lambda & -1 \\ -2 & -\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - \lambda - 2$
 $\Rightarrow \lambda = 2, -1$

- Eigenvalues: $\lambda = 2, -1$

Eigenvalue Decomposition

- To find eigenvector: $(A - \lambda I)x = 0$

- when $\lambda = 2$, $A - \lambda I = \begin{bmatrix} -1 & -1 \\ -2 & -2 \end{bmatrix}$

$$(A - \lambda I)x = 0 \Rightarrow \begin{bmatrix} -1 & -1 \\ -2 & -2 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x = \begin{bmatrix} -1 \\ 1 \end{bmatrix}$$

- when $\lambda = -1$,

$$A - \lambda I = \begin{bmatrix} 2 & -1 \\ -2 & 1 \end{bmatrix}$$

$$(A - \lambda I)x = 0 \Rightarrow \begin{bmatrix} 2 & -1 \\ -2 & 1 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- Eigenvectors: and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

Eigenvalue Decomposition

$$\begin{bmatrix} -1 \\ 1 \end{bmatrix} \quad \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

- $\lambda = 2, -1$ and corresponding eigenvectors are $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$

- The matrix S contains eigenvectors of A in its columns –
 $S = \begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}$

and

- The diagonal matrix Λ contains eigenvalues of A –
 $\Lambda = \begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}$

- $A = \underbrace{\begin{bmatrix} -1 & 1 \\ 1 & 2 \end{bmatrix}}_S \underbrace{\begin{bmatrix} 2 & 0 \\ 0 & -1 \end{bmatrix}}_\Lambda \underbrace{\begin{bmatrix} -2/3 & 1/3 \\ 1/3 & 1/3 \end{bmatrix}}_{S^{-1}} = \begin{bmatrix} 1 & -1 \\ -2 & 0 \end{bmatrix}$

Symmetric matrix

- Symmetric matrix is a square matrix which is equal to its transpose

Example:

$$A = \begin{bmatrix} 4 & 8 & 3 \\ 8 & 5 & 7 \\ 3 & 7 & 1 \end{bmatrix} \quad A^T = \begin{bmatrix} 4 & 8 & 3 \\ 8 & 5 & 7 \\ 3 & 7 & 1 \end{bmatrix}$$

- Symmetric matrices need not be invertible. A matrix with zero entries is symmetric but not invertible

Eigenvalue Decomposition

- A real symmetric matrix can be factored into



$$\mathbf{A} = \mathbf{Q}\mathbf{\Lambda}\mathbf{Q}^T = (\text{orthogonal}) (\text{diagonal}) (\text{orthogonal})$$

- A symmetric matrix has real eigenvalues
- The eigenvectors of a real symmetric matrix are orthogonal to one another provided that they have distinct eigenvalues

Eigenvalue Decomposition

Example:

$$\begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$$

For a matrix $A =$

To factorize A as $A = Q\Lambda Q^T$

- A is symmetric

- To find eigenvalue: $\det(A - \lambda I) = 0$

$$\begin{vmatrix} 3-\lambda & 2 \\ 2 & 3-\lambda \end{vmatrix} = 0 \Rightarrow \lambda^2 - 6\lambda + 5 = 0 \Rightarrow \lambda = 5, 1$$

- Eigenvalues: $\lambda = 5, 1$

• •

Eigenvalue Decomposition

- To find eigenvector: $(A - \lambda I)x = 0$

- when $\lambda = 5$, $A - \lambda I = \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix}$

$$(A - \lambda I)x = 0 \Rightarrow \begin{bmatrix} -2 & 2 \\ 2 & -2 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

- Eigenvectors: and $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

- when $\lambda = 1$,

$$A - \lambda I = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$(A - \lambda I)x = 0 \Rightarrow \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

$$x = \begin{bmatrix} 1 \\ -1 \end{bmatrix}$$

Eigenvalue Decomposition

- For the matrix $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix}$, we have eigenvalues $(\lambda) = 5, 1$ and

eigenvectors $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$

- We may factorize the matrix A into $Q\Lambda Q^T$ as $A = \begin{bmatrix} 3 & 2 \\ 2 & 3 \end{bmatrix} = \underbrace{\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}}_Q \underbrace{\begin{bmatrix} 5 & 0 \\ 0 & 1 \end{bmatrix}}_\Lambda \underbrace{\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ 1/\sqrt{2} & -1/\sqrt{2} \end{bmatrix}}_{Q^T}$

- Q has orthonormal vectors in its columns and Λ has eigenvalues in its columns

Singular Value Decomposition

- SVD – factorization of 1 matrix (any matrix) into 3 matrices

- Any $m \times n$ matrix $\mathbf{A}$ can be factored into
 $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T = (\text{orthogonal}) (\text{diagonal}) (\text{orthogonal})$

- The columns of $\mathbf{U}$ ($m \times m$) are eigenvectors of $\mathbf{A}\mathbf{A}^T$
- The columns of $\mathbf{V}$ ($n \times n$) are eigenvectors of $\mathbf{A}^T\mathbf{A}$

- The diagonal matrix $\mathbf{\Sigma}$ has square roots of eigenvalues from $\mathbf{A}^T\mathbf{A}$ and not from $\mathbf{A}$
- Those positive entries will be $\sigma_1, \dots, \sigma_r$. They are the **singular values** of $\mathbf{A}$ and they fill the first r places in the main diagonal of $\mathbf{\Sigma}$ – when $\mathbf{A}$ has rank r . The rest of $\mathbf{\Sigma}$ is zero

Singular Value Decomposition

$$\begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$$

Example:

For a matrix $A =$

To factorize A as $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$

- To find $\mathbf{U}$:

$$\mathbf{A}\mathbf{A}^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 0 \\ 0 & 4 \end{bmatrix}, \text{ eigenvalues} = 4, 2$$

Eigenvectors of $\mathbf{A}\mathbf{A}^T =$ $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ for the eigenvalue 4 and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ for the eigenvalue 2

- To make the vectors orthonormal, divide them by its length, which will

give $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ and $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$; $\mathbf{U} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$

Singular Value Decomposition

- To find $\mathbf{V}$:

$$\mathbf{A}^T \mathbf{A} = \begin{bmatrix} 1 & 0 \\ 0 & 2 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 4 & 0 \\ 1 & 0 & 1 \end{bmatrix}, \text{ eigenvalues} = 4, 2, 0$$

Eigenvectors of $\mathbf{A}^T \mathbf{A} =$ for the eigenvalue 4, $\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$ for the eigenvalue 2 $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

and for the eigenvalue 0 $\begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}$

- To make the vectors orthonormal, divide them by its length, which will

give , and ; $\mathbf{V} =$

$$\begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} \quad \begin{bmatrix} -1/\sqrt{2} \\ 0 \\ 1/\sqrt{2} \end{bmatrix} \quad \begin{bmatrix} 0 & 1/\sqrt{2} & -1/\sqrt{2} \\ 1 & 0 & 0 \\ 0 & 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$$

Singular Value Decomposition

- The diagonal matrix Σ takes the square root of the nonzero eigenvalues

$$\Sigma =$$

• So, we have,

$$A = \underbrace{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sqrt{4} & 0 & 0 \\ 0 & \sqrt{2} & 0 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} 0 & 1 & 0 \\ 1/\sqrt{2} & 0 & 1/\sqrt{2} \\ -1/\sqrt{2} & 0 & 1/\sqrt{2} \end{bmatrix}}_{\mathbf{V}^T} = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}$$

Singular Value Decomposition

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Example:

For a matrix $A =$

To factorize A as $\mathbf{A} = \mathbf{U}\mathbf{\Sigma}\mathbf{V}^T$

- To find $\mathbf{U}$:

$$\begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$$

$$\mathbf{A}\mathbf{A}^T =$$

; eigenvalues = 3, 1, 0

$$\text{Eigenvectors of } \mathbf{A}\mathbf{A}^T = \begin{bmatrix} -1 \\ 1 \\ 1 \end{bmatrix}$$

for the eigenvalue 3 ,

for the eigenvalue 1

and for the eigenvalue 0

$$\begin{bmatrix} 2 \\ 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix}$$

Singular Value Decomposition

- To make the vectors orthonormal, divide them by its length, which will

give , and

$$\begin{bmatrix} \sqrt{2/3} & 0 & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix}$$

$\Rightarrow \mathbf{U} =$

Singular Value Decomposition

- To find $\mathbf{V}$:

$$\begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$\mathbf{A}^T \mathbf{A} = \quad ; \text{ eigenvalues} = 3, 1$$

Eigenvectors of $\mathbf{A}^T \mathbf{A} =$ $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$ for the eigenvalue 3 and $\begin{bmatrix} -1 \\ 1 \end{bmatrix}$ for the eigenvalue 1

- To make the vectors orthonormal, divide them by its length, which will

give $\begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$ and $\begin{bmatrix} -1/\sqrt{2} \\ 1/\sqrt{2} \end{bmatrix}$ $\mathbf{V} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{2} \\ 1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}$

Singular Value Decomposition

- The diagonal matrix Σ takes the square root of the nonzero eigenvalues

$$\Sigma =$$

- So, we have,
- $$A = \underbrace{\begin{bmatrix} \sqrt{2/3} & 0 & -1/\sqrt{3} \\ 1/\sqrt{6} & 1/\sqrt{2} & 1/\sqrt{3} \\ 1/\sqrt{6} & -1/\sqrt{2} & 1/\sqrt{3} \end{bmatrix}}_{\mathbf{U}} \underbrace{\begin{bmatrix} \sqrt{3} & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}}_{\Sigma} \underbrace{\begin{bmatrix} 1/\sqrt{2} & 1/\sqrt{2} \\ -1/\sqrt{2} & 1/\sqrt{2} \end{bmatrix}}_{\mathbf{V}^T} = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 1 & 0 \end{bmatrix}$$

Complex Numbers?

- Till now we have dealt only with real numbers. What about the complex numbers?
- In real life problems, we mostly deal only with real values and hence we have ignored the complex numbers

Thank you